Analyzing Personal Performance and Patterns in the Rhythm Game osu: Insights from Gameplay Data

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***Abstract*—This study investigates gameplay performance in *osu!*, a rhythm-based game, by analyzing 301 plays recorded by a single participant over a two-month period. Gameplay metrics, beatmap characteristics, and derived performance features such as combo ratio, object density, and miss rate were collected using the *osu!* API and manual logging. Data cleaning included handling missing values, standardizing categorical labels, and ensuring consistency of numerical metrics. Descriptive statistics, univariate and bivariate visualizations, correlation analysis, and statistical tests (t-tests and ANOVA) were conducted to explore patterns and relationships in player performance. Results indicate variations in accuracy, combo ratio, and miss rate across beatmap types and highlight key factors influencing gameplay outcomes. This study demonstrates the feasibility of using personal gameplay data for performance analysis and provides insights for rhythm game performance improvement.**

***Index Terms*—*osu!*, gameplay analysis, rhythm games, data science, performance metrics, combo ratio, object density, miss rate, statistical analysis, visualization.**

I. Introduction

Personal data tracking, often referred to as the "quantified self," enables individuals to gain insights into habits, skills, and performance through systematic logging. Extending this concept to gaming behavior allows players to analyze their own performance, identify patterns, and understand factors influencing success in complex interactive tasks. Rhythm games, such as *osu!*, require precise timing, motor coordination, and sustained attention, and record detailed gameplay metrics, including accuracy, combo counts, and object interactions.

Despite the popularity of rhythm games, few personal longitudinal studies examine individual gameplay performance using detailed play-level data. This study addresses this gap by analyzing 301 personal plays in *osu!*, providing an opportunity to explore how gameplay features and beatmap characteristics influence *Combo Performance Ratio* over time.

The primary research question is: How do gameplay features and beatmap characteristics relate to Combo Ratio in *osu!*? Secondary research questions include:

1) How does Combo Ratio vary across different beatmap types (Hybrid, Jump, Stream, Tech)?
2) What relationships exist between Combo Ratio and other metrics, such as accuracy, maximum greats, and pass/fail outcomes?
3) Do beatmap features or difficulty levels correlate with higher or lower Combo Ratio?
4) Can patterns in Combo Ratio reveal personal strengths or weaknesses across different map types?

The objectives of this study are to collect and analyze all available personal gameplay data, visualize patterns and distributions, conduct statistical tests (e.g., Pearson correlation, ANOVA), and derive practical insights into how gameplay features and performance metrics correlates with combo performance.

II. Literature Review

Rhythm games, such as *osu!*, require precise timing, motor coordination, and sustained attention, making them an ideal context for studying human cognitive-motor performance. Prior studies have demonstrated that regular rhythm game practice can enhance selective attention, processing speed, short-term memory, and sensorimotor coordination [1]–[3]. These findings suggest that rhythm games provide measurable cognitive and motor benefits, making them a suitable domain for experimental investigation.

Large-scale game analytics and telemetry research has examined session-level gameplay metrics, including accuracy, combo counts, and object interactions, to identify patterns in player behavior, engagement, and skill progression [**?**], [4]. Experimental studies, such as Kihara and Hagiwara [1], combined rhythm game sessions with cognitive testing and EEG recordings to assess attentional functions, confirming correlations between gameplay metrics and cognitive outcomes. Other EEG-based and neuroimaging studies similarly indicate that engagement and performance in cognitive games are measurable through neural signals [5], [6]. While these studies provide valuable insights, they primarily rely on aggregated population-level data or short-term laboratory experiments, limiting understanding of longitudinal, individual-level performance.

To address this gap, the present study analyzes 301 plays from a single player, integrating raw gameplay metrics and derived features such as combo ratio, object density, and miss rate. Unlike prior work that emphasizes population-level trends or predictive modeling, this study focuses on personal gameplay data to examine how beatmap characteristics and gameplay features influence individual performance

TABLE I: Dataset Variables, Beatmap Characteristics, and Gameplay Metrics

Variable	Type	Scale / Unit
Metadata and Identifiers		
id	Quantitative	Integer
song_title	Categorical	–
artist	Categorical	–
genre	Categorical	–
beatmap_id	Quantitative	Integer
date_started	Timestamp	YYYY-MM-DD HH:MM:SS
date_ended	Timestamp	YYYY-MM-DD HH:MM:SS
Beatmap Characteristics		
beatmap_type	Categorical	–
difficulty_rating	Quantitative	Stars
bpm	Quantitative	Beats per minute
hit_length	Quantitative	Seconds
count_circles	Quantitative	Count
count_sliders	Quantitative	Count
count_spinners	Quantitative	Count
ar	Quantitative	Approach rate
cs	Quantitative	Circle size
Gameplay Performance Metrics		
accuracy	Quantitative	%
max_combo_player	Quantitative	Count
max_combo_beatmap	Quantitative	Count
max_great	Quantitative	Count
great	Quantitative	Count
ok	Quantitative	Count
meh	Quantitative	Count
miss	Quantitative	Count
is_perfect_combo	Categorical	True/False
passed	Categorical	True/False
played_before	Categorical	True/False

TABLE II: Dataset Variables, Beatmap Characteristics, and Gameplay Metrics with Descriptions

Variable	Description
Metadata and Identifiers	
id	Unique session identifier (Integer)
song_title	Name of the song played
artist	Song artist
genre	Genre of the song
beatmap_id	Unique beatmap identifier
date_started	Timestamp of session start
date_ended	Timestamp of session end
Beatmap Characteristics	
beatmap_type	Beatmap category (Hybrid, Jump, Stream, Tech)
difficulty_rating	Beatmap difficulty rating in stars
bpm	Speed of the beatmap in beats per minute
hit_length	Duration of the beatmap in seconds
count_circles	Number of hit circles in the beatmap
count_sliders	Number of sliders in the beatmap
count_spinners	Number of spinners in the beatmap
ar	Approach rate of the beatmap
cs	Circle size of the beatmap
Gameplay Performance Metrics	
accuracy	Player's accuracy for the session in %
max_combo_player	Maximum combo achieved by the player
max_combo_beatmap	Maximum possible combo of the beatmap
is_perfect_combo	Whether a perfect combo was achieved
passed	Whether the player passed the beatmap
max_great	Maximum great hits for the session
great	Number of great hits
ok	Number of ok hits
meh	Number of meh hits
miss	Number of misses
played_before	Whether the beatmap was previously played by the player

over time, particularly metrics such as *Combo Completion Ratio (CCR)*. By providing detailed, longitudinal insights, this work complements existing rhythm game research while highlighting individual player performance patterns that are typically overlooked in larger-scale analyses.

III. Methodology

A. Participants

The research was carried out with a single participant, who was also the researcher, serving as the sole subject of the study. This participant is a 22-year-old fourth-year student at National University – Manila, pursuing a Bachelor of Science in Computer Science with a specialization in Machine Learning. The participant is an experienced *osu!* player, with prior familiarity in rhythm games, and consistently engages in gameplay sessions spanning various beatmap types to maintain and develop skill.

Throughout the study, gameplay data were self-recorded by the participant over multiple plays, covering a comprehensive range of performance metrics. Data collection was conducted entirely non-invasively, utilizing the *osu!* API in combination with manual logging of session-specific details. No personal or sensitive information beyond the participant's gameplay performance was captured during the process, ensuring that the dataset remained focused strictly on observable gameplay outcomes.

B. Data Collection Methods

Gameplay data were recorded for each session using the *osu!* API, supplemented with manual logging of session-specific details. The dataset was gathered over a one-week period and comprises 302 valid gameplay sessions. Tracked variables include beatmap metadata, beatmap characteristics, and gameplay performance metrics such as accuracy, combo counts, object counts, and hit results. All plays were documented immediately after each play attempt to ensure accuracy and consistency. Only non-sensitive performance metrics were collected, and all recorded data were stored in a Python dataframe for subsequent cleaning, analysis, and visualization. See Table I.

C. Operational Definitions

Each variable in the dataset is defined in Table II:

D. Data Cleaning

Data cleaning was performed to ensure consistency, accuracy, and usability of the dataset for analysis. The following steps were applied:

- **Derived Features:**
  - `combo_ratio` – Calculated as `max_combo_player/max_combo_beatmap`, representing the player's proportion of the maximum possible combo.

- `total_objects` – Calculated as `count_circles + count_sliders + count_spinners`, representing the total interactive elements in a beatmap.
- `object_density` – Calculated as `total_objects/hit_length`, indicating the number of objects per second in the beatmap.
- `miss_rate` – Calculated as `miss/total_objects`, providing a normalized measure of missed notes relative to total objects.

- **Missing Values:** The dataset originally contained missing entries in `meh` (57 missing) and `miss` (3 missing). These were imputed with zeros, as missing hits typically indicate no occurrence rather than a recording error. All other columns were complete.
- **Data Type Standardization:** Numerical features (e.g., `accuracy, max_combo_player, max_combo_beatmap, max_great, great, ok, meh, miss, ar, cs, bpm, hit_length, difficulty_rating, total_objects, object_density, miss_rate, combo_ratio`) retained their original `float` or `integer` types. Boolean metrics (`is_perfect_combo, passed, played_before`) remained as Python booleans. Categorical variables (`genre, beatmap_type`) were stored as strings, with `beatmap_type` standardized.
- **Categorical Standardization:** The `beatmap_type` column was reviewed to ensure consistency. Minor inconsistencies in spelling, capitalization, and spacing were corrected, resulting in four uniform categories: Hybrid, Jump, Stream, and Tech.
- **Temporal Data:** `day` and `hour` were extracted from session timestamps (`date_started`) to allow chronological sorting and temporal analysis.
- **Outliers:** No major outliers were removed, as all values were within plausible ranges. Extreme values (e.g., very high BPM or combo counts) were preserved because they reflect actual gameplay performance.

After cleaning, the final dataset contained 301 complete sessions with 27 columns, including both raw and derived features, ready for exploratory and statistical analysis.

E. Statistical Analysis

Statistical analysis was conducted to explore relationships and patterns in gameplay performance, focusing on `combo_ratio` and beatmap features. The following techniques were applied:

- **Pearson Correlation:** To assess linear relationships between `combo_ratio` and numerical gameplay features such as `accuracy`, `max_great`, `object_density`, and `miss_rate`.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \tag{1}$$

where x_i and y_i are individual observations, and $\bar{x}$ and $\bar{y}$ are the respective means.

- **Mann-Whitney U test:** To compare the distribution of `combo_ratio` between two groups, e.g., different beatmap types or previously played (`played_before=True`) versus new maps. The Mann-Whitney U statistic is computed as:

$$U = \min(U_1, U_2), \tag{2}$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1, \tag{3}$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 \tag{4}$$

where n_1, n_2 are the sample sizes of the two groups, and R_1, R_2 are the sums of ranks for each group. A smaller p-value indicates a statistically significant difference in `combo_ratio` distributions between the groups.

- **One-way ANOVA:** To test for significant differences in `combo_ratio` across multiple beatmap types (Hybrid, Jump, Stream, Tech). Assumes normality and homogeneity of variance.

$$F = \frac{\sum_i n_i(\bar{x}_i - \bar{x})^2/(k-1)}{\sum_i \sum_j (x_{ij} - \bar{x}_i)^2/(N-k)} \tag{5}$$

where k is the number of groups, N is the total number of observations, n_i is the group size, and $\bar{x}_i$ is the group mean.

- All analyses were performed in Python using `pandas` and `scipy.stats` in Google Colab. Statistical significance was evaluated at $p < 0.05$. Visualizations such as histograms, boxplots, scatterplots, and heatmaps were generated to support interpretation of trends and relationships.

IV. Results

This section presents the results of the analysis on 301 osu! gameplay sessions. We report descriptive statistics, visualizations, correlations, and hypothesis-testing results. Each table and figure is accompanied by a title, caption, and a concise interpretation.

A. Dataset Overview

Table III summarizes the numeric features of the dataset, including raw metrics (accuracy, max combo, hits), derived features (combo ratio, object density, miss rate), and temporal variables (day, hour).

TABLE III: Numeric Feature Summary

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
accuracy	301	91.713	5.480	70.230	88.540	92.760	95.820	99.230
max_combo_player	301	353.452	227.062	68	189	293	452	1301
max_combo_beatmap	301	1196.505	572.397	402	745	1093	1520	3273
max_great	301	783.894	352.387	282	503	733	963	1880
great	301	655.625	306.573	86	411	597	821	1747
ok	301	75.668	68.369	2	23	54	112	427
meh	301	7.355	9.608	0	1	4	11	65
miss	301	17.256	13.367	0	7	14	24	68
ar	301	9.126	0.191	8	9	9	9.2	10
cs	301	3.990	0.233	3	4	4	4	5.7
bpm	301	179.167	22.736	87	170	175	190	320
hit_length	301	176.738	73.627	73	113	164	216	441
count_circles	301	462.781	238.625	149	294	422	545	1338
count_sliders	301	319.445	164.734	74	175	287	416	870
count_spinners	301	1.668	1.624	0	1	1	2	8
difficulty_rating	301	5.470	0.401	4.447	5.192	5.452	5.797	6.305
combo_ratio	301	0.333	0.213	0.041	0.172	0.269	0.421	1.000
miss_rate	301	0.022	0.016	0.000	0.010	0.020	0.031	0.100
total_objects	301	783.894	352.387	282	503	733	963	1880
object_density	301	4.443	0.782	2.946	3.971	4.309	4.748	7.667
hour	301	17.754	4.846	0	15	19	22	23
day	301	3.216	2.673	0	1	2	6	8

B. Univariate Analysis

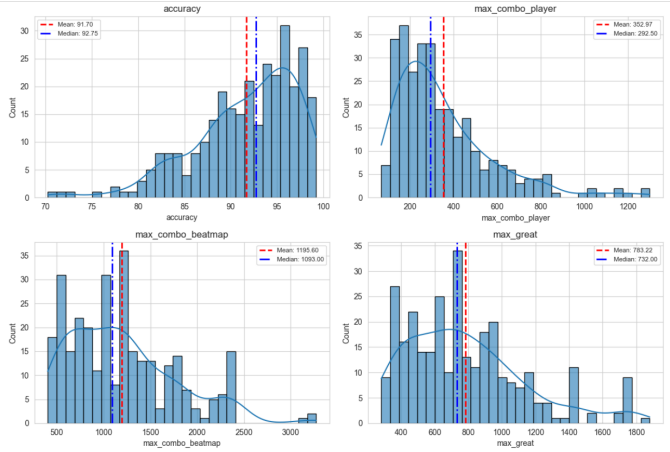

Fig. 1: Histogram distributions of accuracy, max combo player, max combo beatmap, and max great.

Histogram and KDE plots show that accuracy is slightly left-skewed, with most sessions clustering around 92–95%, a mean of 91.70%, and a median of 92.75%, indicating a few low-accuracy outliers. In contrast, max combo player exhibits a right-skewed distribution, with a mean of 352.97 and a median of 292.50, suggesting that most sessions have lower combos while a few high-performance sessions increase the average. Max combo beatmap similarly shows right-skewness, with a mean of 1195.60 and a median of 1093.00, reflecting variability in beatmap design and length. Max great also follows a right-skewed pattern, with a mean of 783.22 and a median of 732.00, indicating that while most sessions achieve moderate "great" counts, some high-performing sessions produce larger values (Fig. 1).

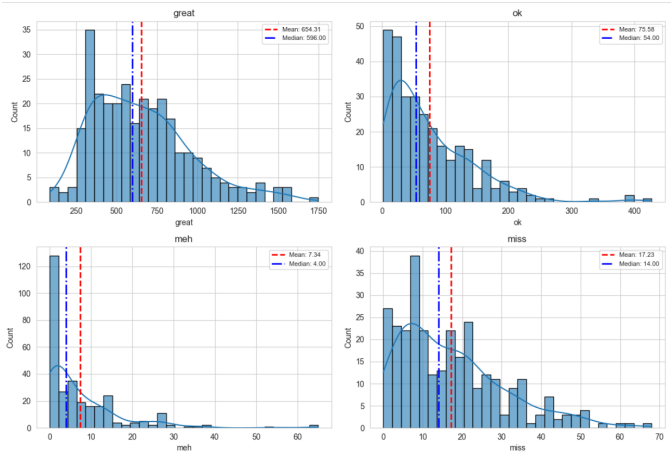

Fig. 2: Histogram distributions of great, ok, meh, and miss.

The distributions of hit accuracy metrics related to note judgments are predominantly right-skewed. The number of "great" hits shows a mean of 654.31 and a median of 656.00, indicating that most sessions cluster near the median with occasional very high counts. The "ok" hits display stronger right skewness, with a mean of 75.58 and a median of 54.00, suggesting that a few sessions contribute unusually high "ok" counts. Similarly, "meh" hits are right-skewed, with a mean of 7.34 and a median of 4.00, reflecting that most sessions have low "meh" counts while some outlier sessions register higher values. The miss count also follows a right-skewed pattern, with a mean of 17.23 and a median of 14.00, highlighting that most sessions have relatively low misses while a few poor-performance sessions increase the average (Fig. 2).

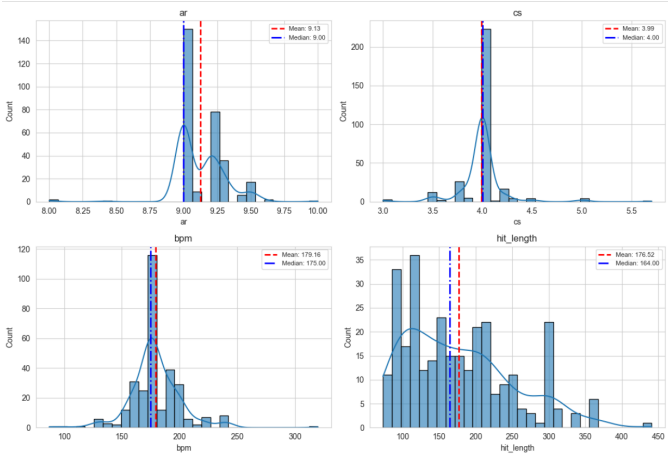

Fig. 3: Histogram distributions of Approach Rate, Circle Size, BPM, and Hit Length.

The distributions of beatmap-related features show varying patterns. Approach rate (AR) is roughly symmetric with a slight left skew, having a mean of 9.13 and median of 9.00, indicating most sessions cluster near the median with few higher-AR maps. Circle size (CS) is approximately symmetric but slightly right-skewed, with a mean of 3.99 and median of 4.00, showing most sessions involve medium-sized circles with occasional larger sizes. Beats per minute (BPM) exhibits a slight right skew, with a mean of 179.16 and median of 175.0, reflecting that most maps have moderate BPM values while some faster maps raise the mean. Hit length is right-skewed, with a mean of 176.52 and median of 164.00, suggesting that while most maps are of moderate length, a few long maps increase the average.(Fig. 3).

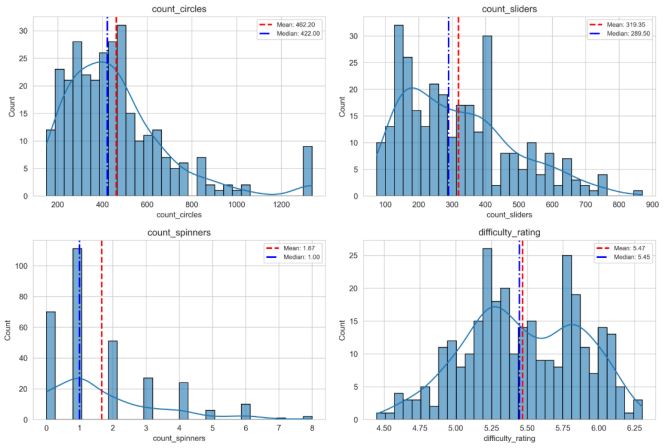

Fig. 4: Histogram distributions of Circle Count, Slider Count, Spinner Count, Difficulty Rating.

The distributions of map object counts and difficulty metrics are generally right-skewed. Circle count has a mean of 462.20 and median of 422.00, indicating most maps contain a moderate number of circles while a few maps with very high counts increase the mean. Slider count is also right-skewed, with a mean of 319.35 and median of 289.50, showing that most maps have a moderate number of sliders but some long maps elevate the average. Spinner count has a pronounced right skew, with a mean of 1.67 and median of 1.00, reflecting that most maps contain few or no spinners while occasional maps include several. Difficulty rating follows a similar pattern, right-skewed with a mean of 5.47 and median of 5.45, indicating that most maps are moderately difficult with some high-difficulty outliers.(Fig. 4).

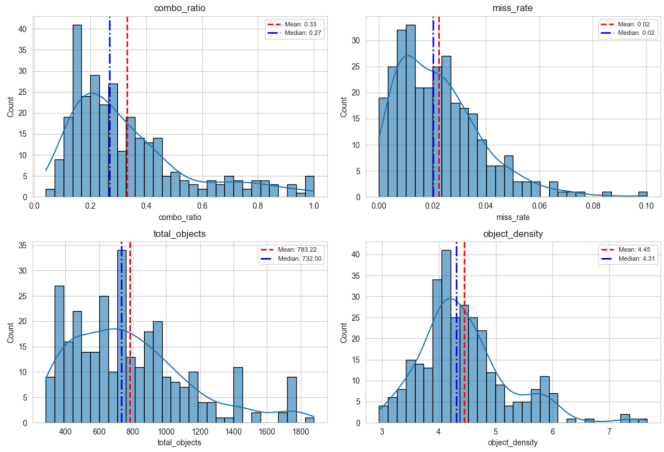

Fig. 5: Histogram distribution of Combo Ratio, Miss Ratio, Total Objects, and Object Density

Performance-related and map composition metrics exhibit right-skewed distributions. Combo ratio has a mean of 0.33 and median of 0.27, indicating that most sessions achieve moderate combo completion while a few sessions reach very high ratios. Miss rate is similarly right-skewed, with a mean and median both around 0.20, showing that most sessions have low miss occurrences but some have unusually high misses. Total objects follow a right-skewed pattern, with a mean of 783.22 and median of 732.00, reflecting that most maps contain a moderate number of objects while larger maps increase the average. Object density also shows a right-skew, with a mean of 4.45 and median of 4.31, suggesting most maps have moderate density but some maps are much more densely packed.(Fig. 5).

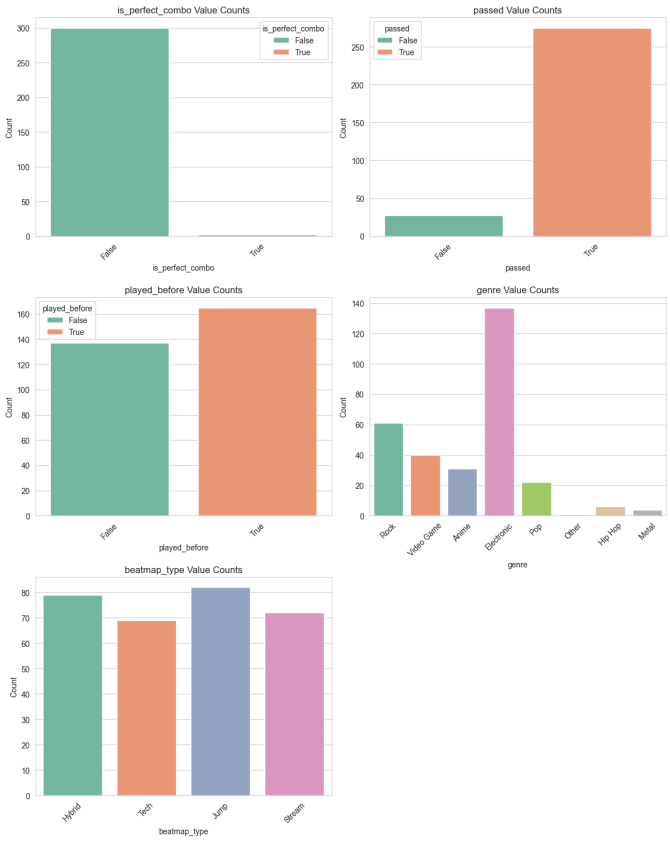

Fig. 6: Histogram distribution of Perfect Combo, Passed, Played, Genre, and Beatmap Type

The Bar Graph shows clear patterns in gameplay outcomes and map selection. For the boolean metrics, almost all sessions were not perfect combos, with only 2 out of 302 sessions achieving a perfect combo. Most sessions were passed (275), while 27 were failed, and the distribution of previously played maps is roughly balanced, with 165 sessions played before and 137 new maps. Among genres, Electronic is the most played (137 sessions), followed by Rock (61) and Video Game music (40), with smaller counts for Anime, Pop, Hip Hop, Metal, and Other. Beatmap types are fairly evenly distributed, with Jump (82) and Hybrid (79) slightly more common than Stream (72) and Tech (69), indicating the player experienced a variety of map types across sessions.(Fig. 6).

C. Temporal Analysis

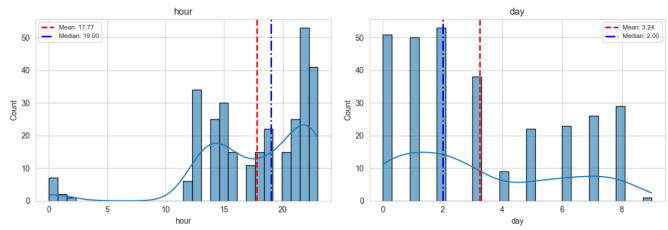

Fig. 7: Histogram distribution of Hours and Days

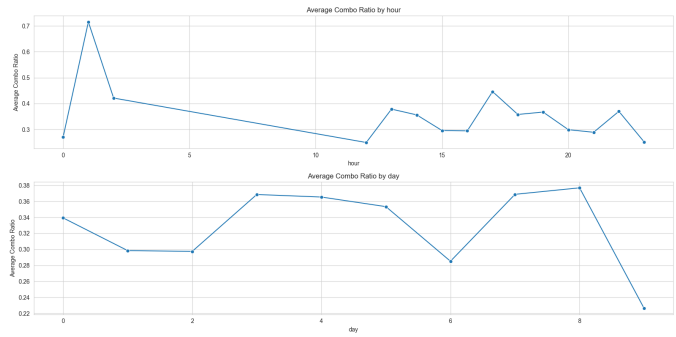

Fig. 8: Line Graph of Average Combo Ratio of Hours and Days

Temporal analysis of gameplay sessions reveals distinct patterns in the timing of play. The distribution of gameplay hours is slightly left-skewed, with a mean of 17.77 and a median of 19.00, indicating that most sessions occurred in the late afternoon to evening, while fewer sessions were recorded in the early morning. Daily session counts show a right-skewed distribution, with a mean of 3.24 and a median of 2.0, suggesting that most days had only a few gameplay sessions, with occasional days of higher activity. These trends highlight the player's preferred time-of-day and variability in daily gameplay frequency. (Fig. 7).

D. Bivariate Analysis

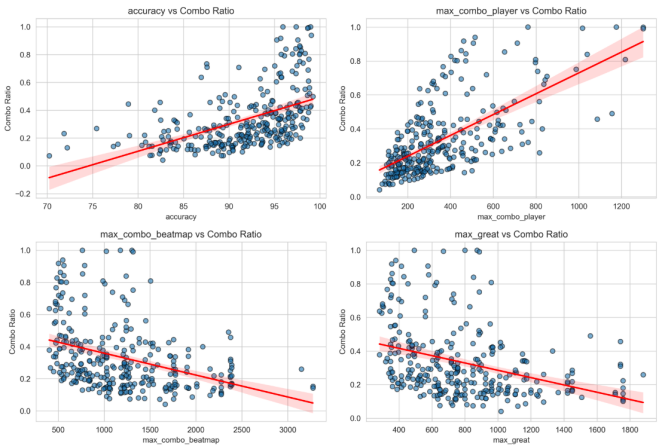

Fig. 9: Scatterplot of accuracy, max combo player, max combo beatmap, and max great.

Scatterplots show that accuracy positively correlates with combo ratio, with higher accuracy generally leading to higher combo ratios. Max combo player also displays a strong positive trend with combo ratio, indicating that sessions with higher maximum player combos correspond to higher performance. Conversely, max combo beatmap and max great show negative trends, suggesting that larger beatmap combos or more great hits in a beatmap do not always translate to higher player combo ratios, potentially reflecting increased map difficulty. (Fig. 9).

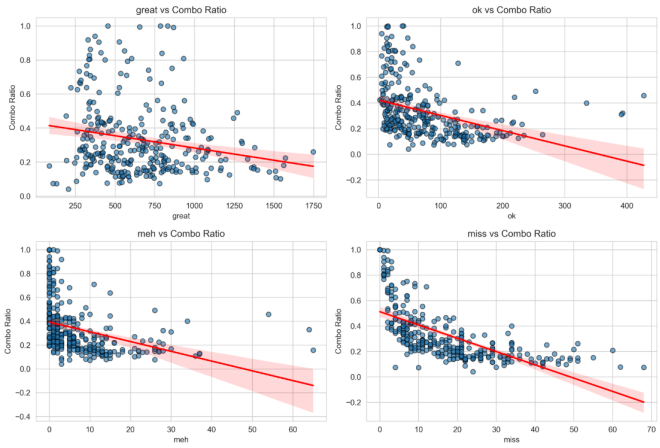

Fig. 10: Scatterplot of great, ok, meh, and miss.

The scatterplots indicate clear negative correlations between combo ratio and the counts of great, ok, meh, and miss hits. Higher numbers of ok, meh, and miss hits correspond to lower combo ratios, while great hits show a mild negative trend, highlighting the inverse relationship between performance errors and sustained combo execution.(Fig. 10).

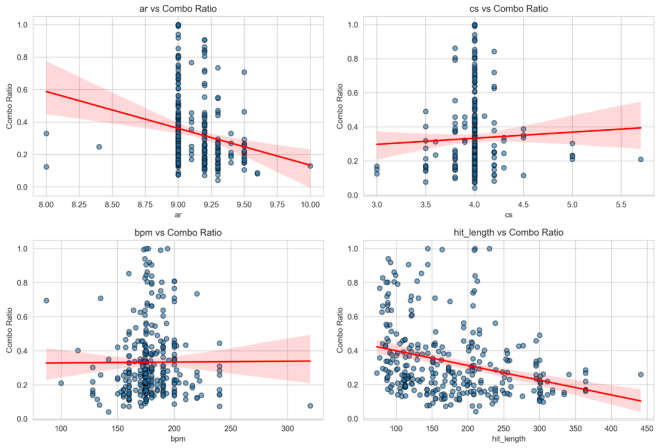

Fig. 11: Scatterplot of Approach Rate, Circle Size, BPM, and Hit Length.

For beatmap parameters, AR shows a slight negative correlation with combo ratio, suggesting that faster approach rates may reduce combo performance. CS and BPM appear largely uncorrelated, with points widely dispersed and the trendline nearly flat. Hit length exhibits a moderate negative trend, indicating that longer songs may be associated with lower combo ratios.(Fig. 11).

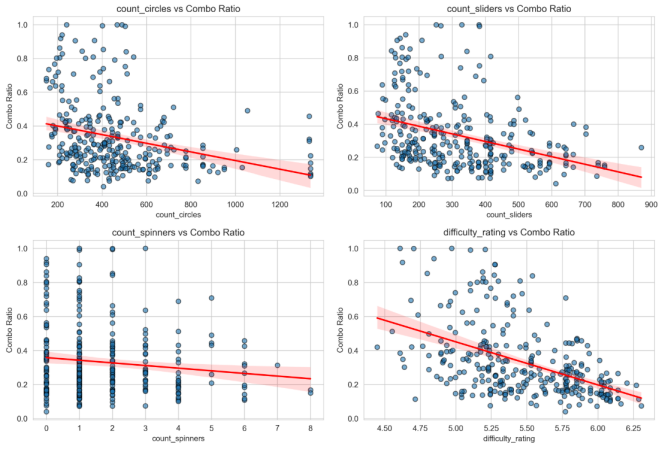

Fig. 12: Scatterplot of Circle Count, Slider Count, Spinner Count, Difficulty Rating.

Scatterplots reveal negative correlations between combo ratio and count_circles, count_sliders, count_spinners, and difficulty_rating. Higher object counts and higher difficulty ratings tend to correspond to lower combo ratios, reflecting the challenge of maintaining long combos on more complex beatmaps.(Fig. 11).

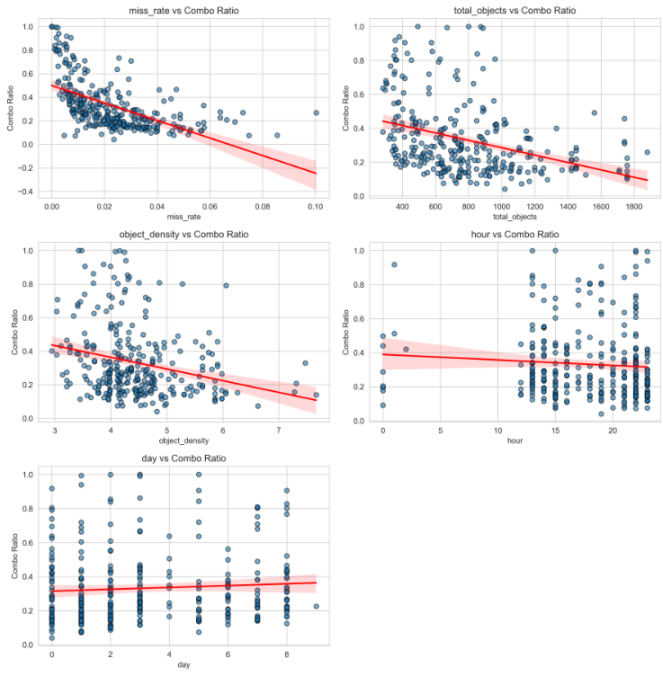

Fig. 13: Scatterplot of Miss Ratio, Total Objects, Object Density, Hours, and Days

The negative trends are pronounced for miss rate, total objects, and object density. Higher miss rates and greater numbers of total objects or object density are associated with lower combo ratios, consistent with the observation that more difficult or dense maps reduce sustained combo performance.(Fig. 13).

Scatterplots show a relatively flat trend between combo ratio and hour, suggesting that time of day does not significantly influence combo performance. Sessions occur mostly in the evening, but performance remains consistent across hours. Combo ratio shows a slight positive trend across days, with no extreme variations, indicating stable performance throughout the dataset timeframe.(Fig. 13).

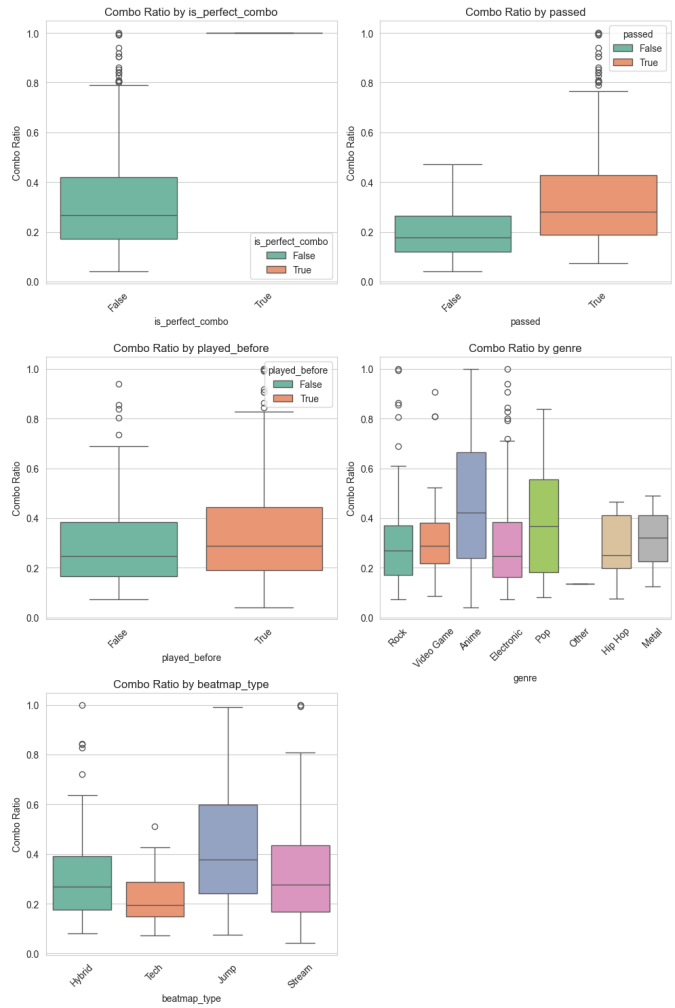

Fig. 14: Boxplot of Perfect Combo, Passed, Played, Genre, and Beatmap Type

Boxplot analysis of categorical and boolean features highlights differences in combo performance across groups. For boolean variables, sessions with a perfect combo (True) show a mean combo ratio of 1.0 and zero variance, while non-perfect sessions (False) have a mean of 0.328 with a standard deviation of 0.206, indicating that perfect combos are rare but maximal when achieved. Passed sessions (True) show a higher mean combo ratio (0.345) compared to failed sessions (False, 0.200), reflecting better overall performance when passing. Similarly, previously played maps (True) exhibit a higher mean combo ratio (0.364) than new maps (False, 0.294), suggesting familiarity improves performance. Genre-wise, Jump maps have the highest mean combo ratio (0.432), followed by Video Game (0.328) and Rock (0.325), while Other shows the lowest (0.136), reflecting differences in player performance across music styles. For beatmap type, Jump maps again show the highest mean combo ratio (0.432), followed by Stream (0.347) and Hybrid (0.314), with Tech maps lowest (0.220), indicating that map design influences combo success. Overall, the boxplots reveal clear group-level differences in performance and

variability, with certain map types, genres, and prior familiarity associated with higher combo ratios.(Fig. 14).

E. Statistical Tests

To examine the relationships between gameplay features and the player's performance, statistical tests were conducted across numeric, boolean, and categorical variables.

TABLE IV: Pearson correlation coefficients for numeric gameplay features.

Feature	r	p-value
miss	-0.656	< 0.0001
max_combo_player	0.655	< 0.0001
miss_rate	-0.564	< 0.0001
accuracy	0.500	< 0.0001
difficulty_rating	-0.477	< 0.0001
ok	-0.382	< 0.0001
meh	-0.370	< 0.0001
max_combo_beatmap	-0.367	< 0.0001
total_objects	-0.360	< 0.0001
max_great	-0.360	< 0.0001
count_sliders	-0.354	< 0.0001
hit_length	-0.300	< 0.0001
count_circles	-0.287	< 0.0001
object_density	-0.256	< 0.0001
great	-0.208	< 0.0001
ar	-0.204	< 0.0001

1) **Pearson Correlation (Numeric Features)**
Pearson correlation coefficients were calculated to assess linear associations between numeric gameplay metrics and performance outcomes, such as `combo_ratio`. Several significant relationships were observed (Table IV). Notably, `miss` showed a strong negative correlation ($r = -0.656$, $p < 0.0001$), indicating that higher misses are associated with lower combo performance. Conversely, `max_combo_player` exhibited a strong positive correlation ($r = 0.655$, $p < 0.0001$), suggesting that higher achieved combos align with better performance. Other metrics, including `miss_rate`, `accuracy`, and `difficulty_rating`, also displayed significant correlations ($|r| > 0.4$, $p < 0.0001$), highlighting their influence on gameplay outcomes.

TABLE V: Mann–Whitney U test results for boolean gameplay features.

Feature	U-stat	p-value
is_perfect_combo	600.000	0.0150
played_before	12,989.500	0.0256
passed	5,339.000	< 0.0001

2) **Mann–Whitney U Test (Boolean Features)**
Non-parametric Mann–Whitney U tests were performed for boolean features to compare distributions of `combo_ratio` across two groups (Table V). Significant differences were found for `is_perfect_combo` ($U = 600.0$, $p = 0.0150$), `played_before` ($U = 12,989.5$, $p = 0.0256$), and `passed` ($U = 5,339.0$, $p < 0.0001$). These results indicate that achieving a perfect combo, prior familiarity with a beatmap, and passing the map are associated with differences in performance outcomes.

TABLE VI: One-way ANOVA results for categorical gameplay features.

Feature	F-stat	p-value
beatmap_type	14.426	< 0.0001
genre	2.721	0.0095

3) **One-Way ANOVA (Categorical Features)**
One-way ANOVA tests were applied to categorical variables to assess differences in `combo_ratio` across multiple groups (Table VI). Beatmap type significantly affected performance ($F = 14.426$, $p < 0.0001$), with certain map types facilitating higher combo ratios. Genre also showed a statistically significant effect ($F = 2.721$, $p = 0.0095$), suggesting that the player's performance varies across different music styles.

These statistical analyses collectively identify which features—numeric, boolean, or categorical—are significantly associated with gameplay performance.

F. Correlation Matrix

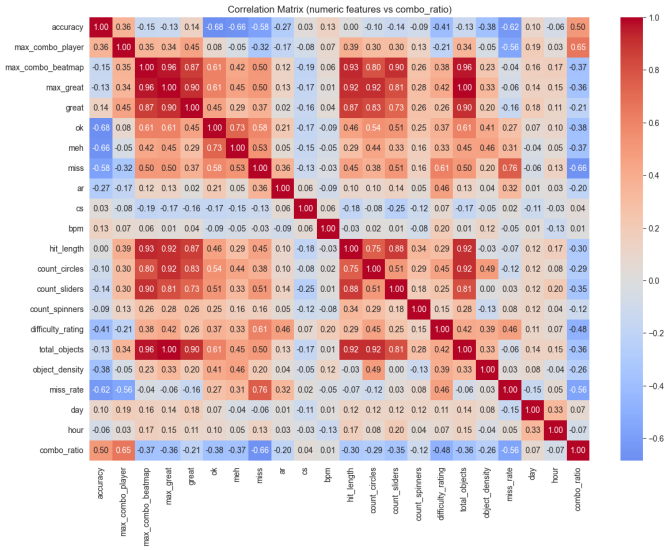

Fig. 15: Correlation Matrix of all numeric features

V. Discussion

VI. Conclusion

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